



HSDP: Hypergraph and structure-aware representation learning for information diffusion prediction

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ABSTRACT

Predicting the micro-level information diffusion is crucial for understanding how content spreads across online platforms, with implications for viral marketing, rumor containment, and recommendation systems. Existing approaches have made progress by leveraging sequential modeling or graph neural networks, yet they often suffer from two limitations: neglecting structural heterogeneity in social networks and failing to capture interactions among cascades. To address these issues, we propose a hypergraph and structure-aware representation learning model for information diffusion prediction (HSDP). It first extracts structural features of users via recursive feature aggregation and encodes them into structure-aware embeddings using a dual graph auto-encoder, enhanced with adjacency reconstruction and contrastive learning to capture the structural heterogeneity. Simultaneously, a hypergraph is constructed to model user participation across cascades and obtain cascade-aware embeddings. We then design a gated mechanism to fuse the two representations, and they are sequentially processed by LSTMs to predict the next adopter. We conduct extensive experiments on four real-world datasets, showing that our model consistently outperforms state-of-the-art baselines on Hits@K and MAP@K scores. The results demonstrate the importance of integrating structural heterogeneity and inter-cascade relationships.

1. Introduction

Predicting how information spreads through online social systems is both a scientifically rich and practically critical problem [1]. It is commonly known as information diffusion prediction or cascade prediction. Users' information-sharing behavior on online platforms forms temporally ordered sequences called cascades. Accurate cascade prediction underlies viral marketing [2], early-warning systems during public emergencies, rumor and misinformation containment, targeted recommendation [3], and epidemiological inference when human contact information is available [4]. The task of cascade prediction (or microscopic prediction) aims to find which user is likely to adopt/re-share an item next, and when. Compared with the macro-level task (final cascade size, known as popularity prediction), microscopic predictions enable interventions such as seeding, throttling, and targeted counter-messaging. The ubiquity of social platforms and the speed at which content propagates make this an urgent applied challenge as well as a long-standing theoretical one [4].

To design accurate, robust, and generalizable cascade predictors, several practical challenges need to be taken into consideration. For

instance, cascades are shaped by both local ties and higher-order group interactions, and in terms of temporal dynamics, cascades are inherently sequential and non-stationary. A rich literature has tackled cascade prediction from multiple angles. Classic methods rely on hand-crafted features that characterize early cascade structures or node attributes, combined with Hawkes or epidemic formulations to model event dynamics [4]. However, these methods share a fundamental limitation that they lack adaptive and self-learning capabilities, preventing them from automatically discovering predictive patterns from data. Feature-based methods require practitioners to manually design a fixed set of descriptors dependent on domain expertise, which are determined prior to training and remain static throughout. Stochastic process models impose pre-specified parametric forms on diffusion dynamics, which introduce strong inductive biases that may be violated in real-world cascades exhibiting heterogeneous and non-stationary behavior. The two classes of approaches both require human-specified intermediate representations that constrain expressive capacity and limit adaptability to diverse diffusion scenarios.

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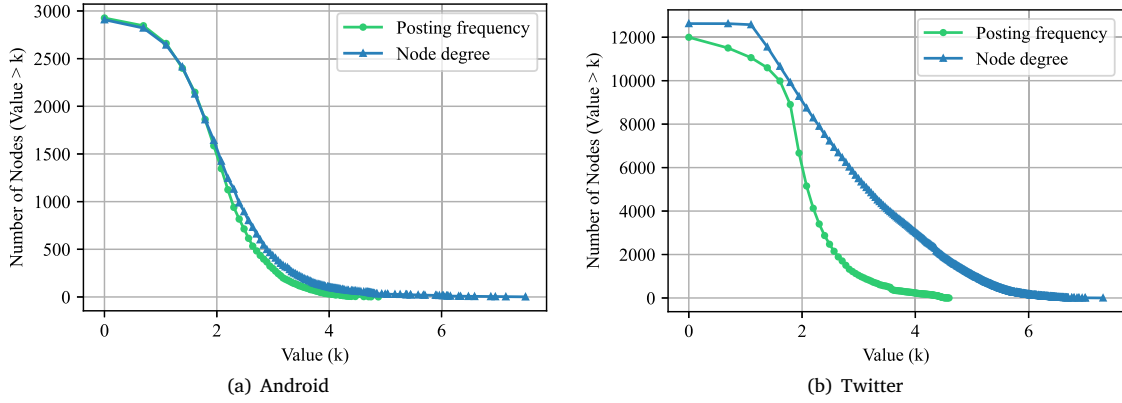


Fig. 1. Complementary cumulative distributions of node degree and posting frequency. The horizontal axis is shown on a logarithmic scale, and the vertical axis represents the number of nodes whose value exceeds a given threshold k .

With the rise of representation learning [5], methods based on end-to-end neural networks have demonstrated superiority in the task of cascade prediction. DeepCas [6] learns a representation of the cascade as a whole via random-walk-based sampling and sequence encoders to predict cascade popularity or growth, avoiding manual feature design. DeepInf [7] and related architectures frame user-level influence as a supervised classification task. They sample a user's local neighborhood and apply GNNs (GCN/GAT) along with attention to predict whether the user will act. These models explicitly exploit the social graph structure around the target node. Topo-LSTM [8] exploits the cascade's diffusion topology by adapting sequence models to topological structures, while neural TPPs and RNN-augmented Hawkes models [1,8] focus on arrival times and capture temporal excitation in cascades. Other methods such as FOREST [9] and Inf-VAE [10] are also based on a similar structure.

Despite substantial progress, there are also some limitations. First, these methods mainly concentrate on a single cascade in the prediction process, but in real-world situations, one cascade can interact with others in various relationships (such as competition and cooperation) [4]. As shown in Fig. 1, we plot the complementary cumulative distributions of node degree and posting frequency on the Android [11] and Twitter [11] datasets. The horizontal axis is displayed on a logarithmic scale, while the vertical axis reports the number of nodes with values greater than k . The green line represents user posting frequency, and we observe that a considerable number of users are simultaneously involved in the diffusion of many messages. Considering that users' attention is limited, it is essential to capture the interactions among cascades in model designs. The underlying driver of such interactions is the co-participation pattern of users across cascades: when a user participates in multiple cascades, their adoption behavior in one cascade is conditioned on their history of participation in others, reflecting mutual influence that single-cascade models inherently ignore.

Second, many existing approaches implicitly or explicitly rely on a pairwise homophily view of diffusion, where influence flows primarily along observed edges and cascades are reduced to sequences or ego-graphs [6,7]. They often overlook structural heterogeneity, where nodes with distinct ego-net shapes or role-like properties (hub nodes, stars, or clique neighborhoods) are treated similarly by homogeneous GNN encoders [12,13]. As shown in Fig. 1, we take the node degree as a brief example, which exhibits similar trends to the posting frequency. It can also be expanded to other complex structural features, indicating that the structural heterogeneity may affect user activity of propagation. Hubs in the social network may be leaders, and they will play different roles in the process of information diffusion in contrast with other leaf nodes. Some methods have been developed for the co-participation pattern of users across cascades, such as CE-GCN [14],

MS-HGAT [15], and DisenIDP [16], but they model inter-cascade relationships in isolation from the structural properties of users in the social network.

Thus, considering the influence of structural heterogeneity on the diffusion process, it might be helpful to generate some structure-aware user embedding for diffusion prediction. Note that we do not discard the notion of social homophily, as it remains an indispensable concept in diffusion prediction. In addition, it is essential to integrate the complex relationships among cascades into the model process, so we propose the hypergraph structure-aware representation learning model for information diffusion prediction (HSDP). Specifically, we first extract structural features of the social network via a simple but efficient method ReFeX [17], which recursively integrates features within neighbors and ego-net, then we generate the structure-aware node embeddings through a GNN auto-encoder framework. To capture relationships among cascades, we construct a hypergraph with all cascades where users participating in a single cascade are represented as a hyper-edge, and generate cascade embeddings via the hypergraph-GNN. These two types of embeddings are fed into two separate LSTMs respectively, and we design a gate strategy to fuse them, which is used for prediction via an LSTM and MLP. Finally, to make the structure-aware embeddings more effective, we add a GNN auto-encoder that reconstructs the adjacency matrix to capture node proximity, and design the contrastive loss to make them become different. The primary contributions of this paper are summarized as follows:

- **Structural heterogeneity modeling from the social network.** We propose to extract structural features of users from the underlying social network via recursive ego-net aggregation. We further design a dual graph auto-encoder module that simultaneously learns structure-aware embeddings and proximity-aware embeddings, regularized by a contrastive loss to enforce complementarity rather than redundancy.
- **Unified integration of structural heterogeneity and inter-cascade hypergraph modeling.** HSDP constructs a hypergraph over all training cascades to model higher-order co-participation patterns, and integrates it with the structural encoder via a gated fusion mechanism. This unification allows simultaneous exploitation of social-role awareness and cross-cascade interaction signals within a single end-to-end framework.
- **Empirical validation on four benchmarks.** We perform extensive experiments on four real-world cascade datasets, demonstrating improved microscopic prediction accuracy and better efficiency compared to state-of-the-art baselines.

The rest of the paper is organized as follows. Section 2 surveys related work on cascade prediction. Section 3 presents the proposed

model in detail, including mathematical formulations, training objectives, and algorithmic procedures. Section 4 describes experiment settings and analyzes model performance. Finally, Section 5 concludes and outlines future research directions.

2. Related work

The task of information cascade prediction has been extensively studied in the past decades. Researchers design models across various methodologies, ranging from traditional models to advanced deep learning approaches. Traditional models often rely on predefined rules to simulate information spread. The Independent Cascade (IC) model [18] assumes that each active node has a single chance to activate its neighbors, with a fixed probability. In contrast, the Linear Threshold (LT) model [19] posits that a node becomes active when the weighted sum of its active neighbors' influence exceeds a certain threshold. Other studies also examined structural diversity of user neighborhoods to quantify the likelihood of resharing, showing that local network structure provides predictive power [20]. Despite their insights, these early models are either too simplistic or heavily rely on expert-driven feature engineering, making them less adaptable to heterogeneous diffusion patterns.

Some studies turn to use probabilistic models to estimate the process of information diffusion. Hawkes processes have been widely used to capture the self-exciting nature of cascades, where each event increases the probability of subsequent adoptions [21], such as DeepHawkes [22] and Reinforced Poisson Process (RPP) [23]. Survival analysis techniques were also adapted for cascade prediction, treating adoption as a time-to-event problem and leveraging hazard-based formulations [24]. These probabilistic approaches provide stronger temporal modeling than regression-based methods, but they often suffer from limited scalability and difficulties in exploiting the underlying graph structure.

With the advent of deep learning, numerous models have been proposed to enhance cascade prediction. Early neural sequence approaches treat a cascade as a timestamped sequence and apply RNNs or attention to encode its temporal order. DeepCas [6] represents a cascade as sampled node walk sequences and uses sequence encoders with attention to predict growth. Topo-LSTM [8] extends LSTM to operate on diffusion topologies so the encoder respects cascade graph structure. DeepInf [7] considers user-level influence as a supervised prediction task and uses graph convolutions and attention over local ego-nets while still relying on sequential cascade signals where appropriate, and Cascade-LSTM [25] designs LSTM variants and generative decoders to predict node-level transmissions and times in cascade trees. Motivated by the success of graph representation learning [26], the expressive power and theoretical foundations of GNNs have been systematically studied from the perspective of graph kernels [12], and recent advances [5,13] have further strengthened the capacity of GNNs to capture structural correspondences across graphs. Several works improve prediction by explicitly learning graph-aware user or cascade representations. Inf-VAE [10] uses VAE-style generative encoders to learn user and cascade latent variables. CasGCN [27] and CasSeqGCN [28] apply graph convolutional networks to cascade or snapshot graphs and combine them with temporal aggregation to capture both structure and dynamics. More recent graph-centric architectures such as CE-GCN [14] extend this line by adding cascade-specific aggregators and cascade-enhanced graph convolutions to better exploit cross-cascade signals.

Many researchers exploit textual and content signals alongside structure. DyDiff-VAE [29] proposes a dynamic variational framework that jointly models evolving user interests and content to estimate propagation likelihoods by combining an encoder of user interest dynamics with a dual-attentive decoder over content and sequence. Some topic-aware models incorporate topic modeling, masked attention, or topic embeddings into the cascade encoder so that content semantics and topical alignment modulate diffusion likelihoods [30,31]. CARE [32] uses

content retrieval and prompt augmentation with social-relation embeddings to combine large-scale pretraining ideas with cascade prediction. Other recent methods model cascades and their relationships as higher-order or heterogeneous objects. MS-HGAT [15] constructs timestamped hypergraphs over cascade participants and applies hypergraph attention to capture higher-order co-participation patterns. Multi-scale and sequential hypergraph approaches [11,33] model cascade interactions via sequential hypergraphs or dynamic hypergraph neural networks to capture cross-cascade competition and cooperation.

Despite significant advancements, existing models often treat nodes uniformly, neglecting structural heterogeneity where nodes possess distinct roles or network motifs. Moreover, many models focus on individual cascades without considering interactions between multiple cascades, such as competition or cooperation. The two most closely related baselines to our work are MS-HGAT [15] and RD-GCN [34], and we clarify the distinctions explicitly here. MS-HGAT constructs multi-scale temporal hypergraphs to capture higher-order co-participation patterns among users across cascades, achieving strong performance through hypergraph attention. However, it treats all users uniformly in terms of their structural roles within the underlying social network. Nodes are encoded identically except through neighborhood aggregation, which is insufficient to capture role-like heterogeneity. RD-GCN incorporates dynamic role information and a gated fusion strategy to combine structural context with temporal diffusion signals. Yet its role features are derived from the constructed diffusion subgraph rather than the intrinsic social network topology, meaning that structural properties such as ego-net density, clustering, and recursive neighborhood patterns, which shape a user's long-term influence capacity, are not explicitly modeled. Furthermore, neither method addresses the complementarity between social-structural features and connectivity proximity, nor do they jointly integrate both inter-cascade hypergraph modeling and social-network structural heterogeneity within a unified framework.

Our proposed HSDP resolves these limitations through three targeted designs. First, we extract structural features from the social network via recursive ego-net aggregation (ReFeX [17]), capturing role-like properties intrinsic to social topology. Second, we design a dual graph auto-encoder with contrastive regularization, which enforces complementarity between structure-aware embeddings and proximity-aware embeddings, preventing redundancy and enriching the representation space. Third, HSDP integrates both the structural heterogeneity module and the inter-cascade hypergraph module in a unified end-to-end framework, enabling simultaneous exploitation of social role awareness and cross-cascade interaction signals.

3. Method

3.1. Problem formulation

We first introduce the problem of information diffusion prediction in social networks. Let the social network be represented as a graph $G = (V, E)$ with adjacency matrix $A \in \mathbb{R}^{N \times N}$, where V denotes the set of nodes/users with $|V| = N$, and E denotes the set of edges with $|E| = M$. For a given dataset $C = \{c_m\}_{m=1}^M$ that consists of M cascades, an information cascade c_m is defined as a temporally ordered sequence of user activations, $c_m = \{(v_i, t_i)\}_{i=1}^k$, where $v_i \in V$ and $t_i \in \mathbb{R}^+$ indicates that user v_i participates in the cascade diffusion process at time t_i . At each time step t , the binary variable $y'_v \in \{0, 1\}$ indicates whether node v is activated. Given the observed cascade $c_m = \{(v_i, t_i)\}_{i=1}^k$ up to time t_k , the task of information diffusion prediction is to estimate the probability that the next inactive node $u \in V \setminus \{v_1, \dots, v_k\}$ will be activated at the next timestamp t_{k+1} , which is denoted as $\hat{y}_u^{t_{k+1}} = \Pr(y_u^{t_{k+1}} = 1 \mid c_m, G)$. In other words, the model is required to predict the future diffusion trajectory by leveraging both the network structure and the cascade sequence.

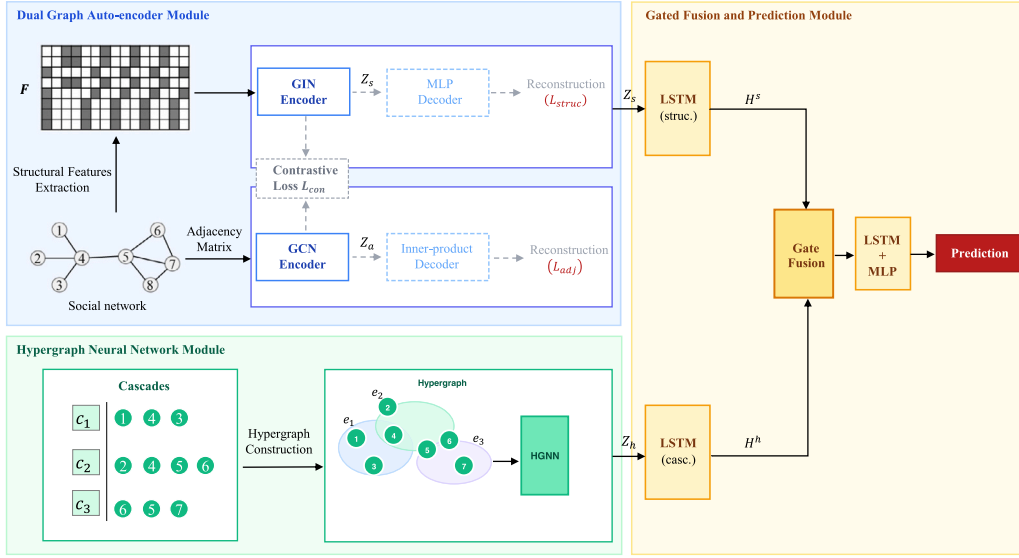


Fig. 2. Illustration of the proposed HSDP model.

3.2. Detail model

As shown in Fig. 2, HSDP consists of three functional modules. We provide a module-level overview before describing each in detail.

The Dual Graph Auto-encoder Module addresses the structural heterogeneity of users in the social network. Users occupying different topological positions in the network, such as hubs, bridges, or peripheral nodes, tend to play distinct roles in diffusion, but most existing methods encode them uniformly through neighborhood aggregation. This module extracts recursive ego-net features from the social graph to characterize each user's structural role, and encodes them into structure-aware embeddings through a GIN-based auto-encoder. In parallel, a GCN-based auto-encoder takes the adjacency matrix as input to produce proximity-aware embeddings that capture pairwise connectivity. A contrastive regularization loss is imposed between the two sets of embeddings to enforce their complementarity and prevent representational redundancy.

The Hypergraph Neural Network Module captures the interactions among cascades that single-cascade models inherently overlook. In real-world platforms, a user's behavior in one cascade is often influenced by their participation in others, reflecting competitive or cooperative dynamics. This module constructs a hypergraph from the training cascades, where each cascade is represented as a hyperedge connecting all its participants. A normalized hypergraph neural network then propagates information between nodes and hyperedges to produce cascade-aware user embeddings that integrate co-participation signals across the entire cascade corpus.

The Gated Fusion and Prediction Module integrates the complementary representations produced by the two preceding modules and generates the final prediction. Since the relative importance of structural and cascade-aware information may vary across different cascades and from different perspectives, a simple concatenation is insufficient. This module employs two view-specific LSTMs to encode the temporal evolution of activated users under each representation, and applies a gated mechanism to adaptively combine their hidden states into unified fused embeddings at each step. The fused sequence is subsequently modeled by a third LSTM, and a multilayer perceptron with softmax outputs the probability distribution over all candidate next adopters.

3.2.1. Hypergraph construction and representation learning

Early studies mostly concentrate on a single cascade and leverage sequential models to predict. However, the user state in one cascade can

be substantially different from that in another, and it may not be appropriate to directly use static user embeddings learned from the social network as inputs to cascade prediction. In other words, a cascade can affect the diffusion process of another through shared participants and overlapping adoption histories, reflecting competitive or cooperative inter-cascade dynamics. Capturing these interactions therefore requires a representation that goes beyond individual cascades and models the relationships among them collectively.

To this end, we introduce a hypergraph structure to jointly represent users and cascades. Unlike conventional graphs that are restricted to pairwise connections, a hypergraph allows a single hyperedge to connect multiple nodes simultaneously, making it well-suited for encoding the set of users that co-participate in the same cascade. Concretely, we construct the hypergraph from the training cascade set and apply a normalized hypergraph neural network that alternates between node-to-hyperedge and hyperedge-to-node message passing, ultimately producing cascade-aware user embeddings that reflect higher-order co-participation patterns across all cascades. For each cascade $c_m = \{(v_i, t_i)\}_{i=1}^k$, we create a hyperedge $e_m = \{v_i \mid (v_i, t_i) \in c_m\}$ that connects all activated users in the cascade. Thus, the hypergraph is defined as $\mathcal{H} = (V, \mathcal{E})$, where V corresponds to the set of users in the social network, and $\mathcal{E} = \{e_m\}_{m=1}^M$ is the set of hyperedges, each representing one cascade. The incidence matrix of the hypergraph is denoted as $\mathbf{H} \in \{0, 1\}^{N \times M}$, where

$$\mathbf{H}_{ij} = \begin{cases} 1, & v_i \in e_j, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

To propagate information over the hypergraph, we adopt a hypergraph neural network (HGNN) to learn user and cascade embeddings [35]. Formally, let $\mathbf{D}_v \in \mathbb{R}^{N \times N}$ denote the diagonal node-degree matrix with $(\mathbf{D}_v)_{ii} = \sum_j \mathbf{H}_{ij}$, and let $\mathbf{D}_e \in \mathbb{R}^{M \times M}$ denote the diagonal hyperedge-degree matrix with $(\mathbf{D}_e)_{jj} = \sum_i \mathbf{H}_{ij}$, where $\mathbf{H} \in \{0, 1\}^{N \times M}$ is the incidence matrix defined in Eq. (1). The normalized hypergraph convolution at the l th layer is then written in two steps.

Step 1: Node-to-hyperedge aggregation. Each hyperedge e_j aggregates the embeddings of its member nodes to produce a hyperedge representation:

$$\mathbf{O}^{(l)} = \sigma \left(\mathbf{D}_e^{-1} \mathbf{H}^T \mathbf{X}_h^{(l)} \mathbf{W}_e^{(l)} \right), \quad (2)$$

where $\mathbf{X}_h^{(l)} \in \mathbb{R}^{N \times d}$ is the node embedding matrix at the l th layer, $\mathbf{W}_e^{(l)} \in \mathbb{R}^{d \times d}$ is a trainable weight matrix, $\sigma(\cdot)$ denotes the ReLU activation function, and $\mathbf{D}_e^{-1}$ normalizes each hyperedge representation by its

degree, preventing large hyperedges from producing disproportionately large aggregated representations.

Step 2: Hyperedge-to-node aggregation. Each node v aggregates the representations of all hyperedges it belongs to:

$$\mathbf{X}_h^{(l+1)} = \sigma(\mathbf{D}_v^{-1} \mathbf{H} \mathbf{O}^{(l)} \mathbf{W}_n^{(l)}), \quad (3)$$

where $\mathbf{W}_n^{(l)} \in \mathbb{R}^{d \times d}$ is a trainable weight matrix, and $\mathbf{D}_v^{-1}$ normalizes each node representation by its hyperedge degree (i.e., the number of cascades the node participates in), ensuring that nodes appearing in many cascades are not unfairly dominated by the sheer volume of aggregated signals. We initialize $\mathbf{X}_h^{(0)}$ with randomly sampled embeddings. After L message-passing layers, we obtain the final user representations $\mathbf{Z}_h \in \mathbb{R}^{N \times d}$, where the i th row corresponds to the cascade-aware embedding of user v_i . The two-step formulation above is equivalent to the unified matrix form:

$$\mathbf{X}_h^{(l+1)} = \sigma(\mathbf{D}_v^{-1} \mathbf{H} \mathbf{D}_e^{-1} \mathbf{H}^T \mathbf{X}_h^{(l)} \Theta^{(l)}), \quad (4)$$

where $\Theta^{(l)} = \mathbf{W}_e^{(l)} \mathbf{W}_n^{(l)}$ is the combined learnable transformation.

3.2.2. Dual graph neural networks for structure-aware node embedding

A key challenge in cascade prediction is that users play heterogeneous structural roles in the social network. Nodes with similar content attributes may behave very differently in diffusion depending on their local topological position. For example, hub nodes tend to act as opinion leaders whose adoptions are more likely to trigger downstream resharing, while peripheral nodes exert limited cascading influence. These structural differences are not captured by conventional neighborhood aggregation, which treats all nodes homogeneously. To address this, we design a dual graph auto-encoder that explicitly encodes structural heterogeneity into user representations.

This module operates through three sequential stages. First, recursive ego-net features are extracted from the social graph to characterize each node's structural role in terms of local connectivity patterns. Second, two parallel auto-encoders are constructed: a GIN-based encoder takes the structural feature matrix as input and is trained to reconstruct it, producing structure-aware embeddings; a GCN-based encoder takes the adjacency matrix as input and reconstructs it via inner products, producing proximity-aware embeddings that reflect pairwise connectivity. Third, because both encoders share the same underlying graph and may otherwise learn redundant representations, a contrastive regularization loss is applied between the two embedding sets to enforce complementarity, ensuring that the structural branch emphasizes role-like topological patterns while the proximity branch captures connection-level similarity.

Structural features extraction. To capture the structural heterogeneity, the extracted features should be sensitive to the local connective pattern of nodes. Here we employ a classical method ReFeX [17], which iteratively aggregates node-level features with those of their neighbors in the ego-net. Specifically, for the node v , the ego-net $\Gamma(v)$ consists of the center node, its neighbors, and all edges among them. We extract one local feature and five ego-net-based features for each node, and the basic features based on $\Gamma(v)$ are defined as follows:

- Number of edges in $\Gamma(v)$: f_1 .
- Sum of node degrees in $\Gamma(v)$: $f_2 = \sum_{u \in \Gamma(v)} d(u)$.
- The proportion of edges in $\Gamma(v)$ to all edges within and leaving $\Gamma(v)$: $f_3 = f_1/f_2$.
- The proportion of edges leaving $\Gamma(v)$ to all edges within and leaving $\Gamma(v)$: $f_4 = (f_2 - f_1)/f_2$.
- The degree of node u : f_5 .
- The clustering coefficient of node u : f_6 .

At each iteration, we augment the feature set of v by aggregating (e.g., mean or sum) features of its neighbors, thereby capturing increasingly larger neighborhoods. After K iterations, each node is represented

by a high-dimensional feature vector $\mathbf{f}_v \in \mathbb{R}^{d_f}$ summarizing its recursive structural patterns. We use $\mathbf{F} \in \mathbb{R}^{N \times d_f}$ to denote the resulting structural feature matrix for all nodes.

Structural graph auto-encoder. While ReFeX provides informative structural descriptors, these features are high-dimensional and potentially redundant. Therefore, we design a graph auto-encoder with a Graph Isomorphism Network (GIN) encoder [36], which can better capture structural information, to distill them into compact structure-aware embeddings. Formally, the encoding layer is given by:

$$\mathbf{h}_{s,v}^{(l+1)} = \text{MLP}^{(l)}\left((1 + \epsilon^{(l)}) \cdot \mathbf{h}_{s,v}^{(l)} + \sum_{u \in \Gamma(v)} \mathbf{h}_{s,u}^{(l)}\right), \quad (5)$$

where $\mathbf{h}_{s,v}^{(0)} = \mathbf{f}_v$, $\epsilon^{(l)}$ is a learnable parameter, and $\text{MLP}^{(l)}$ denotes a trainable multi-layer perceptron. After L layers, we obtain structural embeddings $\mathbf{Z}_s \in \mathbb{R}^{N \times d}$, with row $\mathbf{z}_{s,v}$ corresponding to node v . The decoder is based on the MLP to reconstruct the structural features:

$$\hat{\mathbf{f}}_v = \text{MLP}(\mathbf{z}_{s,v}), \quad (6)$$

with a reconstruction loss defined as the mean squared error (MSE):

$$\mathcal{L}_{\text{struc}} = \frac{1}{N} \sum_{v \in V} \|\mathbf{f}_v - \hat{\mathbf{f}}_v\|_2^2. \quad (7)$$

Adjacency-based auto-encoder. Although the above part captures the structural heterogeneity, it may overlook fine-grained connectivity proximity. To complement this, we employ a second auto-encoder directly based on the adjacency matrix $A \in \{0, 1\}^{N \times N}$. We use a Graph Convolutional Network (GCN) [37] as the encoder:

$$\mathbf{H}_a^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}_a^{(l)} \mathbf{W}_a^{(l)}\right), \quad (8)$$

where $\tilde{\mathbf{A}} = A + I_N$ is the adjacency matrix with self-loops, $\tilde{\mathbf{D}}$ is the diagonal degree matrix, $\mathbf{W}_a^{(l)}$ are trainable weights in the adjacency-based auto-encoder, and σ is a non-linear activation. With $\mathbf{H}_a^{(0)} = I_N$, the encoder outputs adjacency-based embeddings $\mathbf{Z}_a \in \mathbb{R}^{N \times d}$, with row $\mathbf{z}_{a,v}$ corresponding to node v .

The decoder reconstructs the adjacency matrix using the inner product:

$$\hat{A}_{uv} = \sigma(\mathbf{z}_{a,u}^T \mathbf{z}_{a,v}), \quad (9)$$

and the reconstruction loss is the binary cross-entropy:

$$\mathcal{L}_{\text{adj}} = -\frac{1}{N^2} \sum_{u,v \in V} \left(A_{uv} \log \hat{A}_{uv} + (1 - A_{uv}) \log(1 - \hat{A}_{uv}) \right). \quad (10)$$

Contrastive regularization. Directly optimizing the two auto-encoders may lead to redundant embeddings, since both $\mathbf{Z}_s$ and $\mathbf{Z}_a$ derive from the same underlying graph. To enforce complementarity, we introduce a contrastive loss based on cosine similarity. For each node v , we compute:

$$s_v = \frac{\mathbf{z}_{s,v}^T \mathbf{z}_{a,v}}{\|\mathbf{z}_{s,v}\|_2 \cdot \|\mathbf{z}_{a,v}\|_2}. \quad (11)$$

The contrastive loss is:

$$\mathcal{L}_{\text{con}} = \frac{1}{N} \sum_{v \in V} s_v. \quad (12)$$

Minimizing $\mathcal{L}_{\text{con}}$ pushes the two embeddings to be less similar, encouraging one to emphasize the local structure while the other captures the node proximity.

3.2.3. Gate fusion and prediction

The preceding two modules produce complementary user representations: structure-aware embeddings that encode the heterogeneous topological roles of users in the social network, and cascade-aware embeddings that encode cross-cascade co-participation patterns. A naive fusion strategy such as direct concatenation treats both sources of information equally at all times, which is suboptimal given that their relative

informativeness may vary considerably across different cascades and at different propagation steps within the same cascade. Intuitively, cascade-aware embeddings may provide a more reliable signal in the early propagation phase for a single cascade when cascade-specific context is sparse. While structure-aware embeddings could become increasingly informative as the cascade matures and future adoption is more strongly governed by users' structural proximity to the activated subgraph.

Based on the above assumption, we design a gating mechanism that adaptively weights the two views at each propagation step without requiring the transition to be manually specified. Specifically, the two user embedding matrices are fed into two view-specific LSTMs that encode the temporal sequence of activated users in a given cascade and produce per-step hidden states for each view. A gate vector computed from the concatenated hidden states then performs element-wise adaptive weighting to yield fused representations at each timestep. These fused representations are subsequently processed by a third LSTM that summarizes the full cascade history, and a two-layer MLP with softmax activation outputs the predicted probability distribution over all candidate next adopters.

Let $\mathbf{z}_{s,v}$ denote the structure-aware embedding of user v obtained from the dual auto-encoder, and $\mathbf{z}_{h,v}$ denote the hypergraph-based embedding of the same user. Before fusing the two parts of embeddings, we first feed them into view-specific recurrent neural networks to capture temporal states. Given an observed cascade $c_m = \{(v_i, t_i)\}_{i=1}^k$, we feed the sequence of structure embeddings of activated users $\mathbf{x}_i^s = \mathbf{z}_{s,v_i}$ into an LSTM. The per-step updates are:

$$\begin{aligned} \mathbf{i}_i^s &= \sigma(W_i^s \mathbf{x}_i^s + U_i^s \mathbf{h}_{i-1}^s + \mathbf{b}_i^s), \\ \mathbf{f}_i^s &= \sigma(W_f^s \mathbf{x}_i^s + U_f^s \mathbf{h}_{i-1}^s + \mathbf{b}_f^s), \\ \mathbf{o}_i^s &= \sigma(W_o^s \mathbf{x}_i^s + U_o^s \mathbf{h}_{i-1}^s + \mathbf{b}_o^s), \\ \tilde{\mathbf{c}}_i^s &= \tanh(W_c^s \mathbf{x}_i^s + U_c^s \mathbf{h}_{i-1}^s + \mathbf{b}_c^s), \\ \mathbf{c}_i^s &= \mathbf{f}_i^s \odot \mathbf{c}_{i-1}^s + \mathbf{i}_i^s \odot \tilde{\mathbf{c}}_i^s, \\ \mathbf{h}_i^s &= \mathbf{o}_i^s \odot \tanh(\mathbf{c}_i^s), \end{aligned} \quad (13)$$

where $\sigma(\cdot)$ is the sigmoid function, $\odot$ denotes element-wise product, and $W^s, U^s, \mathbf{b}^s$ are trainable parameters. We initialize $\mathbf{h}_0^s = \mathbf{0}, \mathbf{c}_0^s = \mathbf{0}$. The hypergraph view is processed with an analogous LSTM on inputs $\mathbf{x}_i^h = \mathbf{z}_{h,v_i}$, producing per-step states $\mathbf{h}_i^h$.

After the two view-specific LSTMs produce per-step hidden states $\{\mathbf{h}_i^s\}_{i=1}^k$ and $\{\mathbf{h}_i^h\}_{i=1}^k$, we compute a gate vector $\mathbf{g}_i \in [0, 1]^d$ that balances the two embeddings:

$$\mathbf{g}_i = \sigma(W_g[\mathbf{h}_i^s \parallel \mathbf{h}_i^h] + \mathbf{b}_g), \quad (14)$$

where $[\cdot \parallel \cdot]$ denotes concatenation, $W_g \in \mathbb{R}^{d \times 2d}$ and $\mathbf{b}_g \in \mathbb{R}^d$ are trainable parameters, and $\sigma(\cdot)$ is the sigmoid function. The fused embedding $\mathbf{z}_v$ is then obtained as

$$\mathbf{z}_i^{\text{fuse}} = \mathbf{g}_i \odot \mathbf{h}_i^s + (1 - \mathbf{g}_i) \odot \mathbf{h}_i^h, \quad (15)$$

where $\odot$ denotes element-wise multiplication. This adaptive weighting ensures that the prediction process can selectively emphasize structural or cascade-level information depending on the context.

Finally, for the target cascade $c_m = \{(v_i, t_i)\}_{i=1}^k$, the fused user embeddings are fed into another LSTM to evolve the representation of the cascade as

$$\mathbf{h}_m = \text{LSTM}(\{\mathbf{z}_i^{\text{fuse}}\}_{i=1}^k). \quad (16)$$

The final hidden state $\mathbf{h}_m$ summarizes the cascade history up to the k th activation. To predict the next activated user u , we compute the probability of activation for all candidate users $V \setminus \{v_1, \dots, v_k\}$ via a 2-layer MLP:

$$\hat{\mathbf{y}}_u^{t_{k+1}} = \text{softmax}(W_2(W_1 \mathbf{h}_m + \mathbf{b}_1) + \mathbf{b}_2), \quad (17)$$

where $W_*, \mathbf{b}_*$ are trainable parameters. We adopt the cross-entropy loss over all training examples:

$$\mathcal{L}_{\text{pred}} = - \sum_{m=1}^M \sum_{i=1}^{|c_m|-1} \log \hat{\mathbf{y}}_{v_{i+1}}^{t_{i+1}}, \quad (18)$$

where v_{i+1} is the true next activated user in the cascade c_m .

Table 1

Statistics of the four datasets.

Dataset	Nodes	Edges	Cascades	Avg. Length
Christianity	1653	21,955	589	26.02
Android	2927	24,459	679	42.73
Twitter	12,627	309,631	3454	35.21
Memetracker	4709	92,545	12,663	12.31

3.3. Joint training

The final training objective integrates the above components. Since $\mathcal{L}_{\text{struc}}$ and $\mathcal{L}_{\text{pred}}$ serve as primary supervision for structure-aware representation and cascade prediction respectively, we keep them as separate additive terms. The adjacency reconstruction and contrastive regularization serve as auxiliary constraints to enrich the learned embeddings, and thus are weighted jointly by a hyperparameter α :

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{struc}} + \alpha(\mathcal{L}_{\text{adj}} + \mathcal{L}_{\text{con}}). \quad (19)$$

3.4. Complexity analysis

We now analyze the computational complexity of HSDP by considering its major components: the dual graph auto-encoder, the hypergraph neural network, the fusion module, and the sequential prediction module. Let $N = |V|$ denote the number of users, $E = |E|$ the number of social links, $C = |C|$ the number of cascades, L the average length of a cascade, and d the embedding dimension. The dual auto-encoder consists of two parts. For the structural feature reconstruction, the ReFeX process requires $O(E)$ operations to recursively aggregate features, and the MLP decoder has cost $O(Nd^2)$ per epoch. For adjacency reconstruction, the encoder is a GNN layer with complexity $O(Ed)$, while the inner-product decoder requires computing similarities between node pairs. In practice, negative sampling is applied, so the cost is $O(Nd + Ed)$ per epoch rather than $O(N^2d)$.

The hypergraph is constructed with cascades as hyperedges. Each propagation step in the hypergraph GNN involves message passing between nodes and hyperedges, with total complexity $O(CLd)$, where L is the average length of cascades. For each cascade, the LSTM encoder processes L steps. Each step requires $O(d^2)$ operations, yielding $O(Ld^2)$ per cascade. Over all cascades, this is $O(CLd^2)$. The gated fusion adds $O(d)$ per step, which is negligible compared to the LSTM cost. The final prediction module is implemented as a multi-layer perceptron (MLP) followed by a softmax, which costs $O(Nd)$ when N is large.

Combining the above, the total complexity per epoch is approximately

$$O(Ed + Nd^2 + CLd + CLd^2 + CLND), \quad (20)$$

Since in real-world datasets $L \ll N$, and negative sampling reduces quadratic adjacency reconstruction, the model scales linearly with the number of edges E and cascade events CL . Therefore, HSDP is computationally efficient and applicable to large-scale social network datasets.

4. Experiments

4.1. Dataset

To evaluate the effectiveness of our proposed model HSDP, we conduct experiments on four real-world datasets: Android, Christianity, Twitter, and Memetracker. Statistics of the four datasets are reported in Table 1. These datasets cover a wide spectrum of social media platforms and online communities, ranging from microblogging and online forums to thematic discussion groups. The details are as follows:

Table 2
Hits@K (K = 10, 50, 100) scores on Android and Christianity.

Method	Android			Christianity		
	@10	@50	@100	@10	@50	@100
DeepDiffuse	0.0413 $\pm$.0012	0.1058 $\pm$.0021	0.1721 $\pm$.0018	0.1027 $\pm$.0015	0.2183 $\pm$.0024	0.3074 $\pm$.0019
NDM	0.0485 $\pm$.0010	0.1424 $\pm$.0019	0.1897 $\pm$.0022	0.1541 $\pm$.0018	0.3136 $\pm$.0027	0.4586 $\pm$.0025
FOREST	0.0868 $\pm$.0014	0.1773 $\pm$.0023	0.2408 $\pm$.0020	0.2485 $\pm$.0022	0.4201 $\pm$.0031	0.5128 $\pm$.0028
DyHGCN	0.0810 $\pm$.0011	0.1638 $\pm$.0018	0.2309 $\pm$.0017	0.2662 $\pm$.0019	0.4280 $\pm$.0029	0.5247 $\pm$.0026
MS-HGAT	0.0664 $\pm$.0013	0.1648 $\pm$.0020	0.2590 $\pm$.0021	0.2416 $\pm$.0020	0.4550 $\pm$.0033	0.5656 $\pm$.0030
RD-GCN	0.0823 $\pm$.0012	0.1881 $\pm$.0022	0.2786 $\pm$.0019	0.2712 $\pm$.0021	0.4617 $\pm$.0030	0.5756 $\pm$.0027
HSDP	0.0949 $\pm$.0010	0.2004 $\pm$.0018	0.2829 $\pm$.0016	0.2911 $\pm$.0017	0.5334 $\pm$.0025	0.6334 $\pm$.0022

- **Android [11]**: This dataset originates from the Android forum, where users discuss technical topics related to the Android operating system. The social network is formed by user interactions, and cascades correspond to discussion threads initiated by original posts and followed by replies or reposts.
- **Christianity [11]**: This dataset is collected from the Christianity online forum. Users are connected through topic-driven discussions, and cascades correspond to posts and responses under religious topics.
- **Twitter [11]**: This dataset is constructed from retweet cascades on the Twitter social platform. Users correspond to Twitter accounts, and cascades correspond to sequences of retweets triggered by an original post.
- **Memetracker [38]**: This dataset tracks the propagation of textual phrases (memes) across online media outlets and blogs. Each meme spread is treated as a cascade, while the underlying hyperlink network serves as the social graph.

In all datasets, we follow the same preprocessing procedure to ensure fair comparison. We split each dataset into training, validation, and test sets with the ratio of 80%, 10%, and 10%, respectively. To avoid data leakage, the hypergraph construction is performed only on the cascades in the training set. For each cascade, we set the maximum cascade length to 500, truncating sequences that exceed this length. This unified setup ensures consistent evaluation across datasets and prevents information of test cascades from influencing model training.

4.2. Baselines

To evaluate the effectiveness of our model, we compare it with several recent baseline models. The details of the baseline models are as follows:

- **DeepDiffuse [39]**: DeepDiffuse integrates recurrent neural networks with diffusion sequences to predict future activations. It learns representations from temporal cascade structures without relying on hand-crafted features.
- **NDM [40]**: The Neural Diffusion Model leverages temporal point processes and recurrent neural networks to capture the temporal dynamics of diffusion. It directly models the probability of user activation over time, focusing on sequential adoption signals.
- **FOREST [9]**: It constructs diffusion forests to represent the evolution of cascades, where each subtree encodes a diffusion path. It emphasizes structural information in cascade evolution and demonstrates superior performance in modeling diffusion topologies.
- **DyHGCN [41]**: This method explicitly models cascades as dynamic heterogeneous graphs. It employs GCN-based encoders to learn user and cascade embeddings by aggregating temporal and structural signals jointly.
- **MS-HGAT [15]**: It introduces hypergraph modeling to capture higher-order user co-participation patterns. By applying attention mechanisms on multi-scale hypergraphs, it effectively models complex inter-cascade interactions.

- **RD-GCN [34]**: This method incorporates role information of users and models the temporal evolution of diffusion subgraphs. It employs a gated mechanism to combine dynamic role-aware features with structural context, achieving improved cascade prediction performance.

4.3. Experiment settings

All experiments were conducted on a dedicated Ubuntu server equipped with two NVIDIA RTX 3090 GPUs, each with 24 GB memory, and an Intel(R) Xeon(R) Silver 4210 CPU with 256 GB RAM. The software stack was based on Python 3.10 and PyTorch 2.5, with CUDA 12.1 and cuDNN libraries to accelerate GPU computations. Standard machine learning libraries such as NumPy, SciPy, and scikit-learn were also used for data preprocessing and evaluation.

For model training, the batch size was fixed at 32 to balance computational efficiency and gradient stability. We employed the Adam optimizer [42] with an initial learning rate of 0.001, and applied a linear decay scheduler to adjust the learning rate during training. The embedding dimension d was set to 128 for all modules, and the hidden size of all LSTM units matches d . The maximum number of training epochs is set to 50.

The recursive structural feature extraction runs for $K = 3$ iterations. At each iteration, both the mean and sum operators are applied to aggregate neighbor feature values within the ego-net, following the original ReFeX formulation [17]. The resulting structural feature vector after 3 iterations serves as the input to the GIN-based auto-encoder.

As for the neural network depth, both the GIN encoder and the GCN encoder consist of 2 graph convolutional layers. The HGNN likewise uses $L = 1$ message-passing layers. The MLP decoder in the structural auto-encoder and the final prediction MLP each consist of 2 fully connected layers with ReLU activation. All three LSTMs (structural view, cascade view, and final prediction) are single-layer unidirectional LSTMs with hidden size $d = 128$.

For regularization, dropout with a rate of 0.3 is applied in all fully connected layers. Early stopping monitors Hits@10 on the validation set with a patience of 10 epochs: training terminates if no improvement is observed within this window, and the checkpoint achieving the best validation Hits@10 is restored for final evaluation. The hyper-parameters of all baselines are tuned to their best reported performance. All results are averaged over 10 independent runs with different random seeds to ensure statistical robustness.

4.4. Experiment results

We evaluate the effectiveness of our model on the diffusion prediction task with the four real-world datasets. All experiments repeat 10 times and we use the Hits@K and MAP@K with $K = 10, 50, 100$ to measure the performance. The results can be seen in Tables 2–5. Several observations can be drawn.

Our proposed HSDP model achieves the best or second-best performance on both Hits@K and MAP@K metrics across most evaluated settings. On the Android dataset, HSDP improves Hits@100 by approximately 5.0% over the strongest baseline RD-GCN, and on the

Table 3
Hits@K (K = 10, 50, 100) scores on Twitter and Memetracker.

Method	Twitter			Memetracker		
	@10	@50	@100	@10	@50	@100
DeepDiffuse	0.0579 _{±.0014}	0.1080 _{±.0022}	0.1839 _{±.0020}	0.1393 _{±.0016}	0.2650 _{±.0025}	0.3477 _{±.0022}
NDM	0.1521 _{±.0018}	0.2823 _{±.0027}	0.3230 _{±.0024}	0.2544 _{±.0020}	0.4219 _{±.0031}	0.5114 _{±.0028}
FOREST	0.2617 _{±.0021}	0.4007 _{±.0033}	0.4975 _{±.0030}	0.2843 _{±.0022}	0.4641 _{±.0034}	0.5677 _{±.0031}
DyHGCN	0.2718 _{±.0019}	0.4505 _{±.0031}	0.5619 _{±.0028}	0.2890 _{±.0021}	0.4630 _{±.0033}	0.5843 _{±.0030}
MS-HGAT	0.2727 _{±.0020}	0.4663 _{±.0032}	0.5564 _{±.0029}	0.2391 _{±.0019}	0.4688 _{±.0034}	0.5835 _{±.0031}
RD-GCN	0.2795 _{±.0018}	0.4680 _{±.0030}	0.5697 _{±.0027}	0.2903 _{±.0020}	0.4727 _{±.0032}	0.5851 _{±.0029}
HSDP	0.2814 _{±.0016}	0.4728 _{±.0028}	0.5645 _{±.0025}	0.2939 _{±.0018}	0.4778 _{±.0029}	0.5893 _{±.0026}

Table 4
MAP@K (K = 10, 50, 100) scores on Android and Christianity.

Method	Android			Christianity		
	@10	@50	@100	@10	@50	@100
DeepDiffuse	0.0230 _{±.0008}	0.0253 _{±.0007}	0.0256 _{±.0006}	0.0727 _{±.0011}	0.0783 _{±.0010}	0.0784 _{±.0009}
NDM	0.0201 _{±.0007}	0.0222 _{±.0007}	0.0293 _{±.0008}	0.0741 _{±.0010}	0.0768 _{±.0009}	0.0786 _{±.0009}
FOREST	0.0383 _{±.0009}	0.0417 _{±.0009}	0.0426 _{±.0008}	0.1264 _{±.0014}	0.1345 _{±.0013}	0.1358 _{±.0012}
DyHGCN	0.0409 _{±.0009}	0.0440 _{±.0009}	0.0450 _{±.0008}	0.1277 _{±.0013}	0.1350 _{±.0013}	0.1374 _{±.0012}
MS-HGAT	0.0242 _{±.0008}	0.0284 _{±.0008}	0.0297 _{±.0007}	0.0919 _{±.0011}	0.1012 _{±.0011}	0.1028 _{±.0010}
RD-GCN	0.0445 _{±.0009}	0.0501 _{±.0010}	0.0522 _{±.0009}	0.1250 _{±.0013}	0.1331 _{±.0012}	0.1385 _{±.0012}
HSDP	0.0484 _{±.0008}	0.0530 _{±.0009}	0.0542 _{±.0008}	0.1364 _{±.0012}	0.1474 _{±.0012}	0.1488 _{±.0011}

Table 5
MAP@K (K = 10, 50, 100) scores on Twitter and Memetracker.

Method	Twitter			Memetracker		
	@10	@50	@100	@10	@50	@100
DeepDiffuse	0.0587 _{±.0010}	0.0680 _{±.0011}	0.0639 _{±.0010}	0.0814 _{±.0011}	0.0869 _{±.0011}	0.0880 _{±.0010}
NDM	0.1241 _{±.0015}	0.1323 _{±.0016}	0.1430 _{±.0015}	0.1357 _{±.0015}	0.1433 _{±.0015}	0.1446 _{±.0014}
FOREST	0.1460 _{±.0017}	0.1521 _{±.0018}	0.1675 _{±.0017}	0.1437 _{±.0016}	0.1521 _{±.0017}	0.1534 _{±.0016}
DyHGCN	0.1487 _{±.0016}	0.1548 _{±.0017}	0.1658 _{±.0016}	0.1464 _{±.0016}	0.1548 _{±.0017}	0.1563 _{±.0016}
MS-HGAT	0.1418 _{±.0017}	0.1595 _{±.0018}	0.1609 _{±.0017}	0.1111 _{±.0014}	0.1215 _{±.0015}	0.1231 _{±.0014}
RD-GCN	0.1507 _{±.0015}	0.1655 _{±.0016}	0.1669 _{±.0015}	0.1439 _{±.0015}	0.1528 _{±.0016}	0.1543 _{±.0015}
HSDP	0.1524 _{±.0014}	0.1598 _{±.0015}	0.1671 _{±.0014}	0.1485 _{±.0014}	0.1575 _{±.0015}	0.1589 _{±.0014}

Christianity dataset, it yields clear and consistent gains over all baselines across all reported metrics. On the Memetracker dataset, HSDP likewise ranks first on all Hits@K and MAP@K metrics. On the Twitter dataset, HSDP achieves the best performance on Hits@10 and Hits@50, as well as MAP@10 and MAP@100; however, it ranks second to RD-GCN on Hits@100 (0.5645 vs. 0.5697) and MAP@50 (0.1598 vs. 0.1655). We attribute these cases to the characteristics of the Twitter graph: its large scale (12,627 nodes, 309,631 edges) and relatively lower inter-cascade user overlap may reduce the marginal benefit of structural heterogeneity modeling, allowing the role-dynamic approach of RD-GCN to remain competitive on certain ranking thresholds. Overall, HSDP demonstrates strong and broad competitiveness across datasets of varying scales and domains.

Traditional graph-based sequence models such as DeepDiffuse and NDM perform relatively poorly, especially on larger datasets. These methods mainly rely on sequential modeling without fully exploiting structural heterogeneity or inter-cascade dependencies. This leads to suboptimal predictions, particularly when diffusion dynamics are strongly influenced by higher-order structures.

Some recent studies such as FOREST, DyHGCN, MS-HGAT, and RD-GCN explicitly consider structural or cross-cascade relations, and generally outperform simpler baselines. DyHGCN consistently surpasses FOREST on both Twitter and Christianity, which indicates the effectiveness of dynamic graph modeling. MS-HGAT leverages hypergraph attention to capture higher-order user co-participation, achieving competitive performance on Christianity and Twitter. RD-GCN shows strong results across all datasets by integrating role-based user dynamics with gated fusion, validating the importance of structural role modeling in diffusion tasks. However, it still lacks explicit structural information of the social network, which limits its generalization ability on diverse networks.

Table 6
Hits@K scores of ablation study on Android and Christianity.

Method	Android			Christianity		
	@10	@50	@100	@10	@50	@100
w/o struc	0.0743	0.1739	0.2412	0.2103	0.4222	0.5291
w/o hyper	0.0875	0.1981	0.2802	0.2884	0.5265	0.6273
w/o gate	0.0887	0.1973	0.2747	0.2728	0.5178	0.6177
w/o con	0.0899	0.2000	0.2774	0.2728	0.5178	0.6229
w/o (con+gate)	0.0829	0.1946	0.2736	0.2658	0.5156	0.6155
HSDP	0.0949	0.2004	0.2829	0.2911	0.5334	0.6334

Based on the above analysis, the strong performance of HSDP can be attributed to three aspects: (1) its ability to capture structural heterogeneity through dual auto-encoders and contrastive regularization, which allows differentiation of users with distinct local structural patterns; (2) its use of hypergraph neural networks to model complex inter-cascade relations, ensuring that cross-cascade competition and cooperation are effectively integrated; and (3) the gated fusion strategy combined with sequential modeling, which adaptively balances structure-aware and cascade-aware representations over time. Together, these design choices enable HSDP to achieve competitive and often superior diffusion predictions compared with existing methods across diverse datasets and evaluation metrics. The relatively marginal differences on certain Twitter metrics, where HSDP ranks second to RD-GCN, suggest that on very large and sparsely overlapping cascade corpora, the benefit of structural heterogeneity modeling may be partially offset by the advantage of role-dynamic diffusion subgraph features, which motivates further investigation in future work.

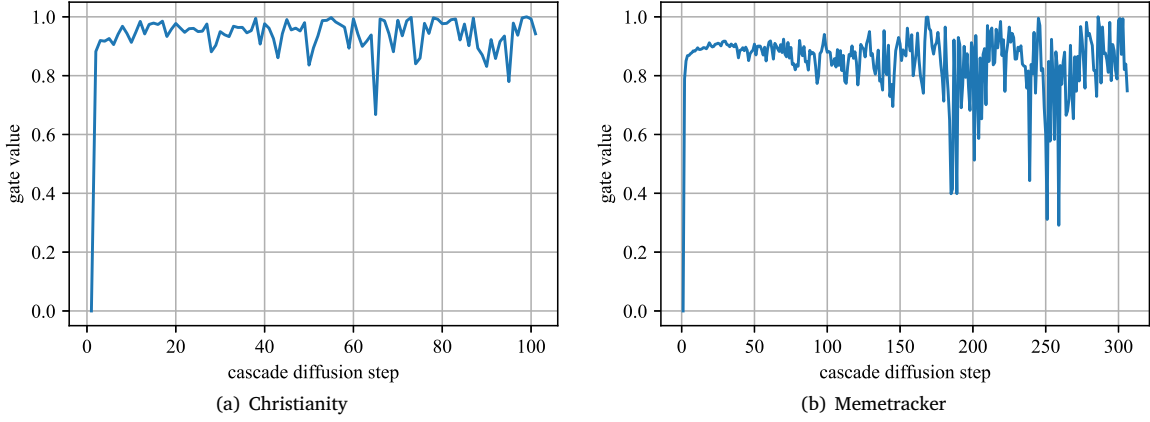


Fig. 3. Mean gate value with cascade diffusion step on the Christianity (left) and Memetracker (right) test sets. Higher values indicate greater reliance on structure-aware representations; lower values indicate greater reliance on cascade-aware representations.

Table 7

MAP@K scores of ablation study on Android and Christianity.

Method	Android			Christianity		
	@10	@50	@100	@10	@50	@100
w/o struc	0.0408	0.0450	0.0459	0.0943	0.1036	0.1051
w/o hyper	0.0461	0.0510	0.0521	0.1362	0.1467	0.1481
w/o gate	0.0470	0.0517	0.0528	0.1296	0.1408	0.1422
w/o con	0.0481	0.0529	0.0540	0.1359	0.1471	0.1486
w/o (con+gate)	0.0459	0.0507	0.0528	0.1272	0.1393	0.1407
HSDP	0.0484	0.0530	0.0542	0.1364	0.1474	0.1488

4.5. Ablation study

To investigate the contribution of each component in the proposed HSDP, we conduct ablation experiments by removing one module at a time: (i) structural representation (**w/o struc**), (ii) hypergraph representation (**w/o hyper**), (iii) gated fusion mechanism (**w/o gate**, directly combining them), (iv) contrastive regularization (**w/o con**), and (v) **w/o (con+gate)**, which retains both the structural and hypergraph branches while simultaneously removing the contrastive loss and replacing gated fusion with direct concatenation. The results on Android and Christianity are reported in [Tables 6](#) and [7](#).

Before presenting the results, we clarify two structural dependencies in the model that are important for correctly interpreting the ablation outcomes. First, the contrastive regularization loss $\mathcal{L}_{\text{con}}$ operates between $\mathbf{Z}_s$ and $\mathbf{Z}_a$, both of which are produced by the dual graph auto-encoder. When the structural encoder is removed (**w/o struc**), the contrastive loss is simultaneously eliminated, as its two operands no longer coexist. Thus, **w/o struc** reflects the combined contribution of structural encoding and contrastive regularization as an integrated branch. Second, when the hypergraph branch is removed (**w/o hyper**), the gated fusion reduces to a trivial pass-through with only one input view. So the performance gap of **w/o hyper** also implicitly subsumes the loss of the gate's adaptive function.

As shown in [Tables 6](#) and [7](#), the ablation results confirm the necessity of each component. **w/o struc** causes the largest degradation across all metrics on both datasets, establishing the structural branch, which jointly encompasses recursive ego-net feature extraction, GIN-based encoding, and contrastive regularization, as the most critical contributor. **w/o hyper** produces a dataset-dependent effect. Its degradation is more pronounced on Christianity than on Android, consistent with Christianity's higher cascade overlap ratio, where inter-cascade co-participation signals carry greater predictive value. **w/o gate** yields consistent drops across all metrics, corroborating the gate weight analysis that the relative importance of the two views shifts across propagation steps and cannot be adequately captured by static concatenation.

w/o con produces the smallest but consistent decrease, confirming that contrastive regularization provides a stable complementarity benefit between $\mathbf{Z}_s$ and $\mathbf{Z}_a$.

The combined variant **w/o (con+gate)** performs worse than either **w/o con** or **w/o gate** individually on all metrics of both datasets, revealing a positive interaction between the two components. Contrastive regularization strengthens the complementarity of the embedding spaces fed into the gate. When both are removed simultaneously, this mutual reinforcement is lost entirely, producing a compounded degradation that exceeds what either removal alone would cause.

To further examine the effectiveness of the gated fusion mechanism, we analyze the behavior of the learned gate weights as a function of cascade propagation step on the Christianity and Memetracker datasets. For each test cascade, we compute the mean gate value $\bar{g}_i = \frac{1}{d} \sum_{j=1}^d (g_i)_j$ at step i , averaged across all test cascades. As shown in [Fig. 3](#), a higher value indicates greater reliance on the structure-aware hidden state, while a lower value indicates greater reliance on the cascade-aware hidden state. On both datasets, the gate value exhibits an overall increasing trend as the cascade diffusion step grows, despite natural fluctuations arising from the varying number and composition of cascades at each step. This trend indicates that the model progressively shifts weight toward the structure-aware representation as more users are activated throughout the diffusion process. The observation is consistent with the design assumption described in [Section 3.2.3](#). These results demonstrate that the gating mechanism acquires meaningful step-dependent adaptive behavior through training, providing a transparent account of how the two representations are prioritized at different stages of cascade propagation.

4.6. Hyper-parameter analysis

We further analyze the influence of the balance parameter α , the embedding dimension d , and the number of iterations K in structural feature extraction on model performance. In our experiments, α is varied from 0.1 to 1.0, d is varied from 2 to 256, and K is in the range [1, 6]. The results are reported on Android and Christianity datasets with Hits@100 and MAP@100 (as shown in [Fig. 4](#)).

- For the parameter α , we observe that the performance is relatively stable within a moderate range. On Android, the best value of Hits@100 appears near $\alpha = 0.7$ – 0.8 , suggesting that auxiliary objectives contribute most effectively when their weights are balanced with the main prediction task. On Christianity, the results are more sensitive: Hits@100 increases steadily from $\alpha = 0.1$ to around $\alpha = 0.5$ – 0.6 (0.642), before slightly declining. MAP@100 follows a similar trend, peaking at $\alpha = 0.9$. This indicates that

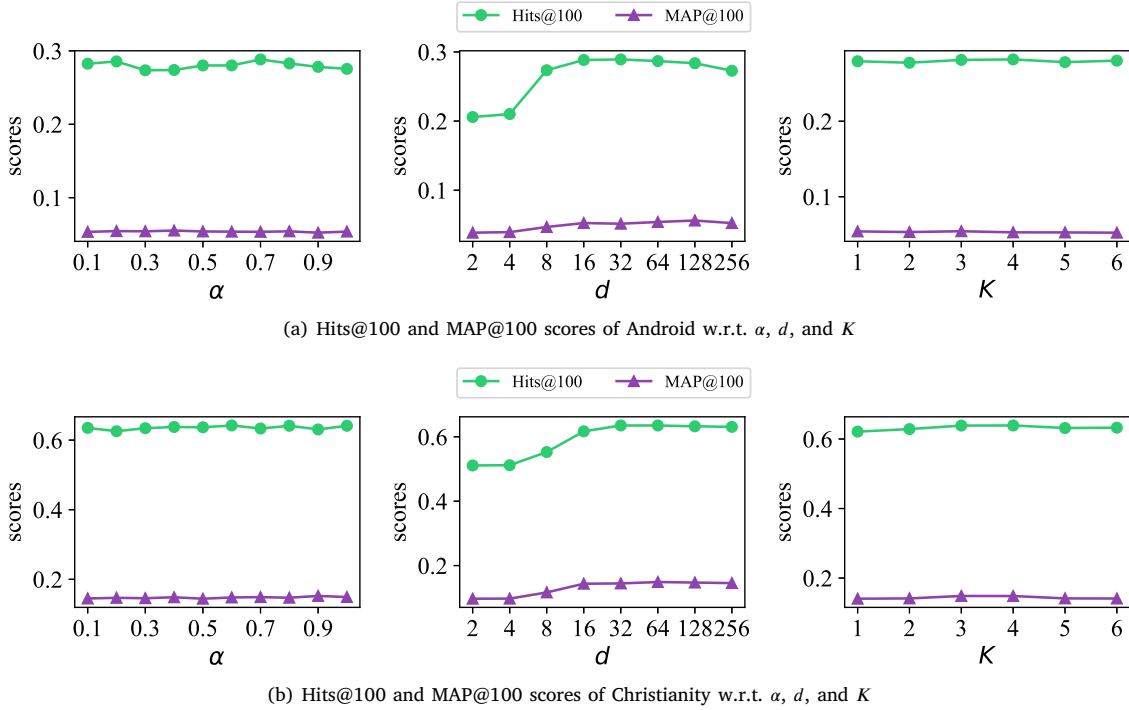


Fig. 4. Diffusion prediction results with different hyper-parameters.

while auxiliary tasks consistently enhance structural embeddings, an excessively large α may cause over-regularization and reduce predictive accuracy.

- For the embedding dimension d , the results reveal that increasing the dimension generally improves performance up to a certain point, after which the gain diminishes or even slightly decreases. On the Android dataset, as d increases, performance steadily improves, peaking around $d = 128$, after which gains saturate and even slightly decline at $d = 256$. A similar pattern is observed on Christianity. These results suggest that medium-sized embeddings (64–128) are sufficient to balance expressiveness and generalization, while higher dimensions may introduce redundancy or overfitting.
- For the number of iterations in the structural feature extraction, as shown in the third sub-panel of Fig. 4, HSDP is overall robust to the choice of K on both datasets. On Christianity, performance peaks in the range $K = 2$ to 4, suggesting that two to four iterations of recursive aggregation provide sufficient structural context. When $K > 4$, performance slightly declines as increasingly large neighborhoods introduce noise from structurally distant nodes. On Android, the curve remains nearly flat throughout, indicating low sensitivity to aggregation depth on this dataset. So $K = 3$ could be a reliable and efficient default choice.

4.7. Efficiency analysis

To empirically validate the scalability implied by the above analysis, we compare the total training time of HSDP against all baselines on the Android and Twitter datasets, as shown in Fig. 5. HSDP achieves competitive running time relative to other GNN-based methods on both datasets, and scales moderately as the graph size increases from Android to Twitter, confirming that the dual auto-encoder and hypergraph module introduce no prohibitive computational overhead in practice.

5. Conclusion

In this paper, we proposed a hypergraph structure-aware representation learning model for information diffusion prediction. Our design was motivated by two key challenges in cascade modeling: the heterogeneity of user structural roles in the social network, and the interdependence among multiple cascades. The proposed model addresses these challenges by introducing a dual graph auto-encoder to generate structure-aware node embeddings, complemented with adjacency reconstruction and contrastive learning to ensure diversity, as well as a hypergraph neural network to capture higher-order cascade interactions. A gating mechanism adaptively fuses these embeddings, which are then modeled sequentially with LSTMs for accurate microscopic prediction. Extensive experiments on four benchmark datasets demonstrated that our model achieves superior performance compared with strong baselines. The improvements are consistent across both small-scale and large-scale datasets, validating the superiority and effectiveness.

Despite these strengths, there are also several limitations of our approach. First, the hypergraph in HSDP is constructed statically from the full training cascade set, and does not account for the temporal evolution of inter-cascade relationships. Cascades that emerge later in time may interact differently from earlier ones, and a continuous dynamic hypergraph formulation could be better. Second, the structural features extracted via ReFeX are based on the observed social network topology, which may be incomplete or noisy in practice due to unobserved links or platform-specific privacy constraints. Third, while HSDP demonstrates competitive efficiency, the dual auto-encoder introduces additional memory overhead that may become a bottleneck for extremely large-scale social networks with millions of nodes. Future work will explore continuous inter-cascade construction to capture temporal dependencies, investigate more robust structural feature extraction under incomplete graph observations, and extend the framework to large-scale industrial settings through more scalable approximations of the auto-encoder components. Incorporating content-level signals alongside structural and cascade-level representations also constitutes a promising direction for further improving prediction accuracy.

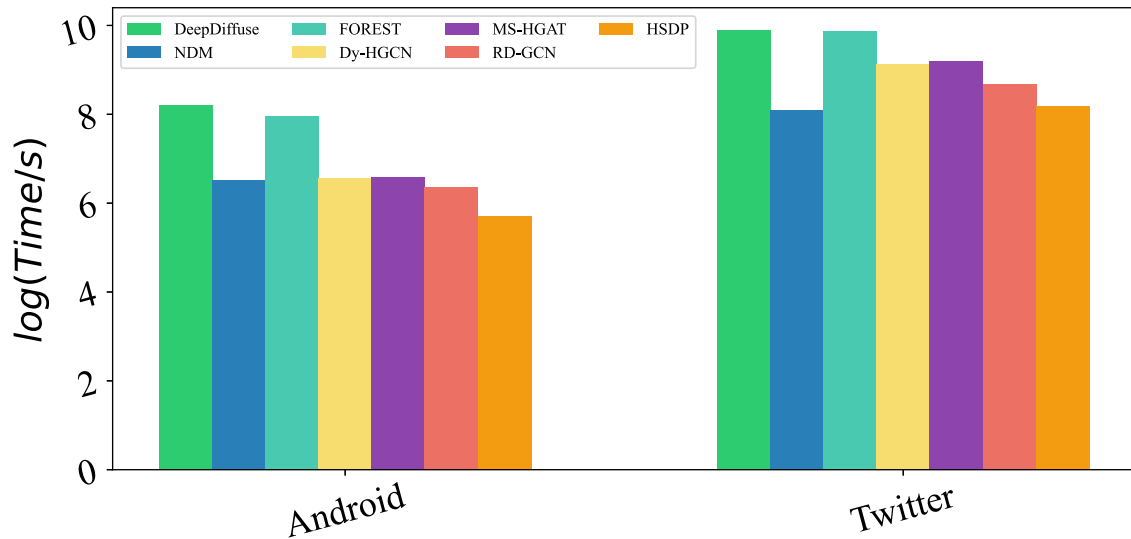


Fig. 5. Total training time (log scale) of all methods on Android and Twitter datasets.

CRedit authorship contribution statement

Wang Zhang: Writing – original draft. **Wenjun Wang:** Resources. **Xuan Guo:** Formal analysis. **Tianpeng Li:** Methodology. **Minglai Shao:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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